Hospital Financial Model Documentation

Purpose

This document explains the structure, calculation methodology, assumptions, and limitations of the Hospital Financial Model. The workbook presents a 25-year operating and project-finance forecast for a 300-bed hospital.

Workbook structure

Section	Purpose
Assumptions	Central input area for hospital capacity, utilization, revenue yield, growth, margin, and debt terms
Sources & Uses	Reconciles project funding with construction, fees, and contingency costs
Revenue	Forecasts annual revenue based on occupied bed days and annual growth
Operating Cost	Calculates operating expense and EBITDA using the operating-margin assumption
Cash Flow	Presents EBITDA as CFADS in the current model structure
Debt Schedule	Calculates annual opening debt, principal, interest, and closing debt
DSCR	Calculates CFADS divided by total debt service
Scenarios	Stores downside, base, and upside assumptions; the base case is selected

Calculation methodology

Revenue

Year 1 revenue is calculated as:

$$\text{Beds} \times \text{Occupancy} \times \text{Days per Year} \times \text{Revenue per Bed Day}$$

Subsequent years are calculated as:

$$\text{Revenue}_t = \text{Revenue}_{t-1} \times (1 + \text{Revenue Growth})$$

Operating costs and EBITDA

The operating margin is assumed to be 18.0% of revenue:

$$\text{EBITDA} = \text{Revenue} \times \text{Operating Margin}$$

$$\text{Operating Expense} = \text{Revenue} - \text{EBITDA}$$

Debt schedule

Debt is initially \$140.0 million and amortizes evenly over 25 years:

$$\text{Annual Principal} = \frac{\$140.0 \text{ million}}{25} = \$5.6 \text{ million}$$

Interest is calculated using the beginning-of-year debt balance:

$$\text{Interest}_t = \text{Beginning Debt}_t \times 5.25\%$$

Debt service coverage ratio

The workbook uses EBITDA as CFADS. DSCR is calculated as:

$$\text{DSCR} = \frac{\text{CFADS}}{\text{Principal Repayment} + \text{Interest Expense}}$$

Sources and uses

The project is funded with \$140.0 million of debt and \$60.0 million of equity. These sources fund \$180.0 million of construction cost, \$10.0 million of fees, and \$10.0 million of contingency. Total sources equal total uses at \$200.0 million.

Model conventions

- Forecast period: 25 annual periods, labeled Years 1 through 25.
- Currency: U.S. dollars, unless the project instructions specify otherwise.
- Revenue: Calculated from average occupied bed days and revenue per occupied bed day.
- Margin: Held constant at the selected scenario assumption in the current model.
- Debt: Fully amortizing, with fixed annual principal and a 5.25% interest rate applied to beginning debt.
- Scenario selection: The workbook currently uses the base scenario—78% occupancy, 3% revenue growth, and 18% operating margin.

Limitations and required checks

The model is a simplified project-finance forecast and should not be treated as a complete hospital valuation, accounting forecast, or credit-underwriting model without further work.

- Verify that the scenario-selection cells flow through every forecast line, rather than serving only as a reference table.
- Confirm formulas, cell links, number formats, and debt-service calculations after any input change.
- Add explicit assumptions for taxes, maintenance capex, working capital, depreciation, cash reserves, and financing fees if CFADS or valuation is required.

- Add historical operating information and externally supported market, utilization, reimbursement, and cost assumptions for a real-world use case.
- Review whether the opening DSCR of 1.31x satisfies the applicable lender covenant or project requirement.